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The science of psychedelic medicine

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Classic psychedelics typically act at the serotonin 5-HT_{2A} receptor to profoundly alter brain function and consciousness. Research on these compounds has accelerated. Major strides have been made in understanding their unique mechanisms of action and clinical potential. This Review outlines the state of psychedelic science, spanning cellular mechanisms, systems neuroscience and clinical investigation. We show that preclinical and human research findings converge on two complementary processes: acute neural desynchronization, which destabilizes entrenched network patterns, and subacute neuroplasticity, which opens a window for psychological and behavioral change. We review evidence of therapeutic response across neuropsychiatric indications and consider how this integrates with mechanistic findings. We also explore challenges and opportunities, including discrepancies between preclinical evidence that non-hallucinogenic psychedelic analogs engage putative therapeutic mechanisms, and clinical evidence linking the subjective experience to therapeutic response; the risks inherent to enhanced neuroplasticity; and questions surrounding trial design, scalability and regulatory approval. The growth of psychedelic science and medicine may compel a fundamental rethinking of the relationship between subjective experience and biological change in psychiatry.

Once relegated to the fringes of psychiatry, psychedelic compounds are now at the forefront of translational neuroscience. This resurgence is disrupting not only psychiatry but also our understanding of how the mind, brain and experience interact[1]. Psychedelic-containing plants and fungi have been used for millennia in Indigenous ceremonial, healing and spiritual practices. However, the ascent to center stage in Western society has been propelled by a growing body of rigorous clinical trials demonstrating therapeutic potential of psychedelics[2]. Clinical trials with psychedelics have demonstrated rapid and sustained improvements in depression[3–7], anxiety[8–12], post-traumatic stress disorder (PTSD)[13] and addiction[14] following just one or two psychologically supported psychedelic sessions. Phase 3 programs are underway for psilocybin in depression, lysergic acid diethylamide (LSD) for anxiety disorder and novel or modified psychedelics for multiple indications.

This wave of positive clinical trials accompanies a growing infrastructure for psychedelic research and treatment around the world. A growing number of academic institutions have set up dedicated psychedelic research programs, training initiatives and clinical units to study and deliver psychedelic treatments. Companies offering psychedelic retreats have grown exponentially[15,16]. At the same time, investment from both established pharmaceutical companies and a new generation of biotech innovators has accelerated discovery of novel analogs, formulations and digital tools for psychedelic-assisted therapy[17–19]. Together, these forces have transformed psychedelics from a niche curiosity into a prominent frontier in translational neuroscience.

Despite this exuberance, researchers remain divided on several foundational questions: whether translational findings in animals can truly model human psychedelic effects, whether subjective

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Table 1 | Hallucinogens and psychoplastogens in psychiatry

Drug class	Receptor target(s)	Example molecules	Drug effects	Therapeutic potential
Hallucinogens				
Classic psychedelics	5-HT_{2A} agonists (although many are nonselective)	• Ayahuasca • DMT • LSD • Mescaline • Psilocybin • 5-MeO-DMT	• Anxiety/paranoia • Ego dissolution • Emotionality • Nausea • Tachycardia • Visual hallucinations	• Depression/ existential distress • Anxiety • Addiction • Headache • Pain
Dissociative anesthetics (or simply 'dissociatives')	Selective NMDA receptor antagonists	• Dextromethorphan • Ketamine • Methoxetamine • Nitrous oxide • PCP	• Analgesia • Dissociation • Hallucinations • Impaired motor coordination • Sedation	• Addiction • Depression • Pain
	Polypharmacological compound	• Ibogaine (also engages opioid, 5HT and DA systems)	As above, plus dreamlike and ego-dissolving	• Opioid addiction • Traumatic brain injury
Empathogens/entactogens (substances that promote emotional openness, empathy and connectedness)	Presynaptic monoamine release • Serotonin • Dopamine • Norepinephrine	• MDMA • Methylone	• Empathy/connection to others • Enhanced sensation • Euphoria • Tachycardia • Diaphoresis • Cognitive impairment during the comedown	• PTSD
Oneirogens (substances that promote dreamlike states)	Kappa opioid receptor agonism	• Salvinorin A (*Salvia divinorum*) • Enadoline • RB-64 • U-50488	• Hallucinations • Dizziness • Dysphoria	None
Non-hallucinogenic 'psychoplastogens'				
NHPAs	5-HT_{2A} partial/biased agonism	• Lisuride • Tabernanthalog • 2-Br-LSD • AAZ-A-154 • ITI-1549	Unknown	• Addiction • Mood • PTSD
Other psychoplastogens	• AMPAr • HDAC • mGluR2/3 • PDE4 • Sigma-1 • TrkB • Wnt	• Ibudilast • Lithium • Pomaglumetad • Roflumilast • Rolipram • 7,8-DHF • CHIR99021 • LM22A-4 • LY354740 • LY379268 • SB-216763	Variable	• Cognitive enhancement • Mood • Neuroprotection

AMPAr, α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptor; DA, dopamine; HDAC, histone deacetylase; 5-MeO-DMT, 5-methoxy-*N*,*N*-dimethyltryptamine; mGluR2/3, metabotropic glutamate receptor 2/3; NMDA, *N*-methyl-D-aspartate; PCP, phencyclidine; PDE4, phosphodiesterase-4; Sigma-1, sigma-1 receptor; TrkB, tropomyosin receptor kinase B (NTRK2); Wnt, wingless/integrated signaling pathway.

experiences are necessary for therapeutic benefit and how to manage expectancy and functional unblinding. Acknowledging these uncertainties is essential to ensure the field advances on a foundation of rigorous science. Our goal is to lay out progress in the neuroscience and clinical science of psychedelics, with an eye toward these unanswered questions and controversies.

A major scientific objective in the field has been to understand the biological underpinnings of the acute state of altered consciousness. This psychedelic state is often characterized by terms such as 'ego dissolution', a temporary destabilization of the coherent sense of self. Classic psychedelics such as psilocybin and LSD produce this effect through the serotonin 5-HT_{2A} receptor[20]. Yet adjacent drug classes can produce similar effects through different mechanisms (Table 1). To complicate matters further, terms such as 'ego' and 'self' lack precise neuroscientific definition and explanation. This makes it difficult to specify how psychedelics disrupt these constructs or to determine whether ego dissolution is a necessary mediator of therapeutic benefit. Thus, connecting subjective experiences to therapeutic mechanisms remains a critical area of investigation as well as contention. These acute effects of psychedelics, however, are only part of the mechanistic story.

Clinical observations that psychedelics loosen rigid patterns of thinking and behavior have been buoyed by a growing body of animal research suggesting that psychedelics activate neuroplasticity pathways[21–25]. Yet the work of translating from rodents to humans and definitively linking neural plasticity to psychological and clinical change remains incomplete. In humans, administration of psychedelics is associated with persistent changes in numerous psychological traits and functions. However, the therapeutic effects of psychedelics are remarkably context sensitive to factors such as expectancy, environment and therapeutic rapport[26]. These factors may amplify both positive and negative influences, raising both opportunities and risks for enduring therapeutic benefit. This Review integrates empirical observations from preclinical, human imaging and clinical research, emphasizing two complementary frameworks: acute neural desynchronization (which ties more broadly to 'entropy', defined below) and persistent neural plasticity (which ties more broadly to 'changeability'). We explore strengths and limitations of these two frameworks, how they are being leveraged in clinical care, and how they may guide next-generation therapeutic development. We explore challenges and controversies, including the case for non-hallucinogenic psychedelic

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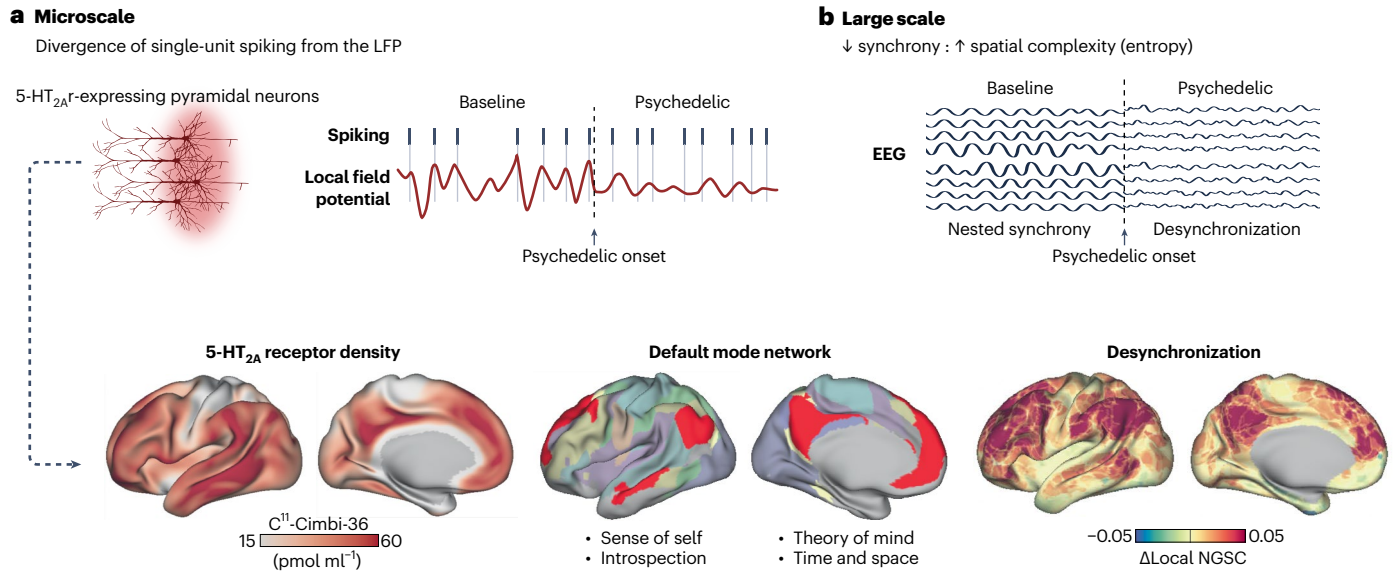

Fig. 1 | Psychedelics induce brain desynchronization. The 5-HT$_{2A}$ receptor is expressed primarily in the association cortex (bottom left), with the highest levels in the DMN (bottom middle). **a**, 5-HT$_{2A}$-receptor-expressing pyramidal neurons are illustrated at the top left and LFP of those neurons is shown to the right. Under normal conditions (baseline), single-unit depolarizations (vertical bars) occur primarily when the LFP is in the active state (for example, bars align with peaks in LFP trace). After a psychedelic, 5-HT$_{2A}$-receptor-expressing neurons desynchronize from the LFP[36]. **b**, Desynchronization is also observed at the large scale or macroscale. A cartoon EEG trace demonstrates nested synchronized oscillations at baseline that become desynchronized after a psychedelic. Below, a map of desynchronization (normalized global spatial complexity or NGSC) across the human cerebral cortex. MRI-based measures of desynchronization correspond to the DMN and correlate with subjective psychedelic effects. Reproduced from ref. 51, Springer Nature Limited.

analogs (NHPAs)—both as pharmacotherapies and as tools for unpacking the complex relationships between the inimitable psychedelic experience and clinical efficacy. By mapping the translational terrain from receptor to behavior, we aim to explore the promise and perils of using psychedelics as tools to catalyze therapeutic change.

Acute desynchronization

Stability and adaptation

The human brain maintains a delicate balance between stability and adaptability[27–29]. Stability supports the construction of a predictive model of reality, allowing us to navigate the present and anticipate the future. Adaptability enables the incorporation of new information and learning. During normal waking consciousness, ensembles of interconnected neurons fire in synchronized and organized rhythms[30,31]. This synchronized activity, nested within large networks, forms a stable scaffolding for brain activity. It is thought to encode a predictive model of the world that constructs 'reality' from the inside out[32], supporting physiological and behavioral habits that are crucial to survival. These synchronized and predictable rhythms also help the brain to distill the torrent of incoming information into a workable picture of reality, letting us predict what will happen next.

But when the patterns become rigid, the mind can get stuck in harmful cycles of thought that impact emotion and behavior, resulting in conditions such as depression, anxiety and addiction[33,34]. Psychedelic drugs appear to give a brief shake-up to entrenched patterns of brain activity; they disrupt established rhythms, which may open a 'window' of time during which the brain becomes more flexible and able to lay down healthier patterns[35].

Empirical observations across scales

Psychedelics seem to loosen the timing and coordination of brain activity at every level of organization. At the microscale (individual neurons), psychedelics decouple cortical pyramidal neuron spiking and the oscillatory phase of local field potentials (LFPs). Fine-grained recordings show that activating the 5-HT$_{2A}$ receptor makes these neurons less likely to fire during the peak of the local field wave and more likely to fire at random moments[36,37] (Fig. 1a). At the level of local neural networks (the 'mesoscale'), studies across species report a broad drop in the strength and synchrony of neuronal population (ensemble) activity and brain rhythms[38]. Human electroencephalogram (EEG) readouts mirror this pattern, with clear reductions in alpha (~8–12 Hz) power and smaller drops in theta (~4–8 Hz) and beta (~13–30 Hz) band power, indicating disruption to cortical rhythmicity[39–42].

At the brain-wide scale (the 'macroscale'), psychedelics disrupt the stable scaffolding; EEG, magnetoencephalography and functional magnetic resonance imaging (fMRI) all reveal global desynchronization and higher signal complexity following administration of psychedelics (Fig. 1b). These observations have collectively been described as an increase in entropy; that is, disorder or randomness in a system[43,44]. These increases in entropy reflect a transition toward a more disordered state of information processing. It is hypothesized that this breaking up of tightly organized and nested cortical rhythms by psychedelics may relax the assumptions that usually shape perception and thought[45]. In Bayesian terms, psychedelics may temporarily suppress activity that maintains priors, opening a brief window in which certain patterns of neural activity and thought become more adaptable. The rise in entropy is, therefore, not just 'noise'; it may be the very mechanism that enables behavioral flexibility[46]. Psychedelic-induced increase in brain entropy appears to predict beneficial changes in both psychological insight and mental health outcomes[47].

Network-specific effects and subjective experience

These disruptive effects of psychedelics are not uniformly distributed across the brain. Instead, psychedelics preferentially disrupt activity in 5-HT$_{2A}$-receptor-rich areas, such as the default mode network (DMN) and visual system[48] (Fig. 1). The DMN is a collection of interconnected brain areas which, in normal consciousness, encode our sense of space, time and self[49]. Psychedelic-induced ego dissolution and altered time perception correlate with desynchronization of activity in hubs of the DMN[50,51], whereas psychedelic visual imagery tracks with disruption in the visual cortex[50,52]—supporting the link between disrupted modular functioning and phenomenology of drug effects[53].

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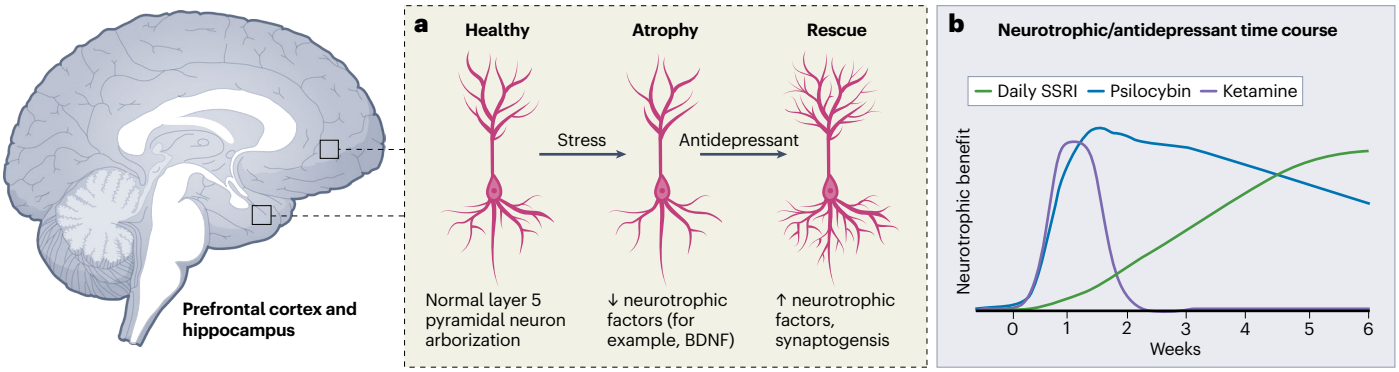

Fig. 2 | Persistent neurotrophic and therapeutic effects. a, Chronic stress leads to suppression of trophic factors such as BDNF and atrophy of pyramidal neurons in the PFC and hippocampus, which correlates with depression-like behavior in preclinical models. Antidepressant drugs tend to increase trophic factor expression and recover synapses lost as a result of stress. **b**, These neurotrophic effects are seen rapidly with ketamine and psychedelics, whereas they take weeks with SSRIs.

Desynchronization as a trigger for neural plasticity

In the desynchronized state, groups of neurons that normally fire together become out of sync with one another, and novel ensembles may emerge[54]. This perturbation is the hallmark of a brain and mind perceiving new or unexpected information or ideas. For example, a sensory stimulus may induce deviations in synchronized activity within perceptual systems, particularly when it is unexpected[55]. This deviation from expectations triggers neural processes that promote plasticity and learning, sometimes referred to as homeostatic plasticity[56].

Psychedelics may represent a pharmacological disrupter, triggering homeostatic plasticity and widespread network recalibration[51]. Similar principles seem to explain the mechanism of action of other psychiatric drugs, such as ketamine (dissociative anesthetic) and lithium (mood stabilizer)[57]. In plain language, the dramatic departure from a stable set point that is brought on by a psychedelic may cause brain activity and behavior to become more plastic and, therefore, adaptable, even after the acute effects have resolved. It was recently shown that the pattern of acute, drug-evoked spiking activity is predictive of subsequent circuit rewiring[58]. Further work is needed to test this proposed link between desynchronization, plasticity and clinical benefit, but also to explore secondary implications, such as whether the therapeutic action of a psychedelic may decrease with sequential psychedelic doses, as the drug effects become less novel.

Persistent plasticity

The neurotrophic hypothesis of depression

The neurotrophic hypothesis of depression proposes that mood disorders result from stress-induced reductions in synaptic plasticity, neurogenesis and structural remodeling[59,60]. Animal models consistently show that chronic stress reduces hippocampal expression of trophic factors and synapse density in the prefrontal cortex (PFC) and hippocampus[60]. Conversely, trophic factor expression, neuritogenesis and synaptogenesis are stimulated by multiple classes of antidepressant drugs and interventions[61–66]. Neurotrophic stimulation appears to be both necessary and sufficient for recovery of depression-related behaviors in animal models which typically index motivational loss (for example, immobility on the forced swim test) and anhedonia (for example, decreased sucrose preference)[67]. Although mostly demonstrated in rodents, the validity of these ideas is also supported by studies showing volumetric loss and atrophy in the hippocampus and PFC in individuals with depression[68]. Yet the implications of neurotrophic stimulation to brain function at the meso, macro and psychological levels is not well understood.

Psychedelics as psychoplastogens

The neurotrophic hypothesis was formalized after increases in brain-derived neurotrophic factor (BDNF) and other neurotrophins were observed weeks after initiating daily dosing with a selective serotonin reuptake inhibitor (SSRI) in animal models[59,64,66]. A decade later, the striking discovery was made that, unlike the gradual effects of SSRIs, ketamine induced a rapid neurotrophic cascade—evident within hours of dosing[61,69–72] (Fig. 2). This followed the demonstrations of rapid antidepressant benefits of subanesthetic ketamine[73]. Across species, chronic stress is well known to cause cortical atrophy[59,60]. The rapid recovery of dendritic spines and synapses lost as a result of stress[74,75] plays a critical role in sustaining the long-term antidepressant-related behavioral effects of ketamine in rodents[74,75]. Therefore, it seems plausible that synaptic recovery might also underlie the therapeutic action of psychedelics in humans.

In preclinical research, psychedelics such as psilocybin and LSD have been shown to activate similar neuroplasticity-related pathways to ketamine, suggesting that their therapeutic effects may stem from an analogous mechanism[76]. A rapid increase in expression of plasticity-related immediate-early genes in the hippocampus and frontal cortex[23,25] leads to BDNF-dependent activation of the TrkB receptor, followed by dendritic spine growth[21,77] and synaptogenesis[22]. Synaptogenesis is seen within 24 h of psychedelic dosing[78], and can persist for more than a month[24] (Fig. 2).

Efforts are ongoing to find reliable in vivo biomarkers for neurotrophic stimulation in humans. Direct assessment—via positron emission tomography of molecular targets in the neurotrophic cascade[79,80] or changes in synaptic density[81–85]—may be possible within the next few years. Initial attempts to measure psychedelic-induced increases in synaptic density with the positron emission tomography tracer ^{11}C-UCB-J have been inconsistent[86,87]. Measurement of plasticity-related brain changes with EEG and fMRI may be more immediately feasible[88]. Peripheral blood biomarkers provide a complementary, indirect window into molecular pathways associated with neuroplasticity[89]. It may also be possible to measure epigenetic modifications in circulating cells that reflect transcriptional programs relevant to plasticity[90,91]. In parallel, platelet-based measures of G-protein membrane localization and cAMP signaling are another promising indicator of neurotrophic signaling and antidepressant response[92,93].

Development of these and other methods to measure neuroplasticity in the human brain in vivo will enable research to answer a range of previously unapproachable scientific questions: do neuroplastic mechanisms underlie psychiatric treatments' effectiveness? Do plasticity signatures differ across effective treatments for depression, PTSD, addiction or other psychopathologies and across

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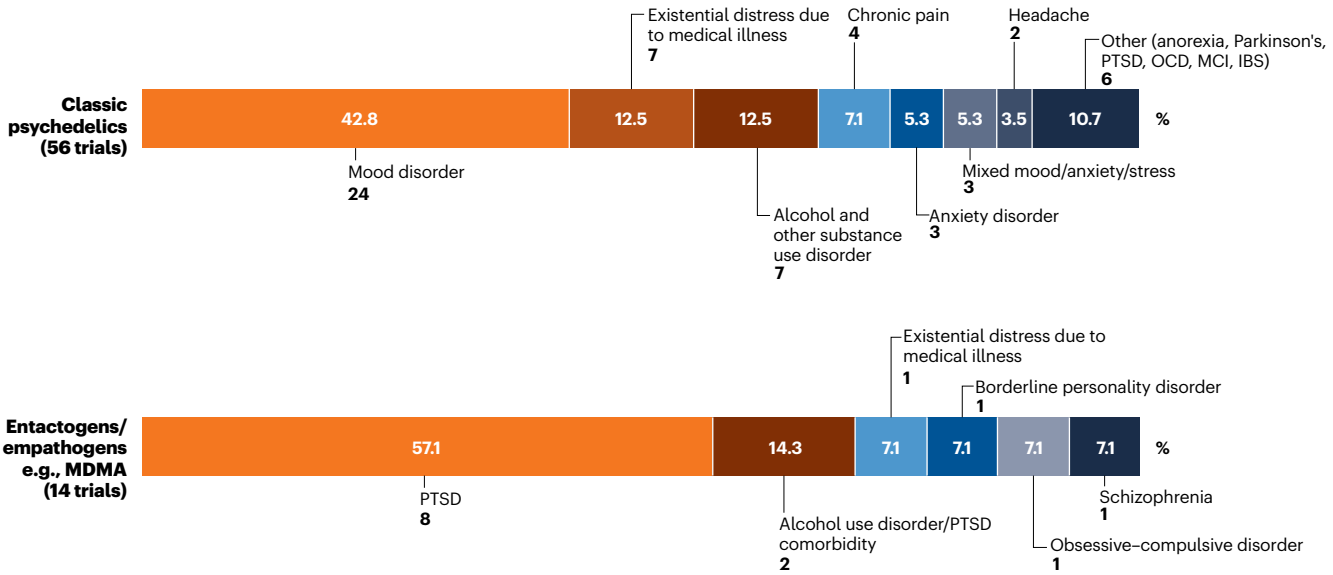

Fig. 3 | Active phase 2/3 clinical trials, grouped by primary clinical condition. Data were extracted from ClinicalTrials.gov on 2 December 2025. A query for classic psychedelics revealed 59 active and currently recruiting clinical trials; 56 trials assessed effects on clinical symptoms. A query for empathogens (such as MDMA and methylone) revealed 14 active and currently recruiting clinical trials. Mood disorder includes MDD, treatment-resistant depression and adjustment disorder. Conditions with only a single trial of classic psychedelics are grouped as Other (PTSD, mild cognitive impairment (MCI), anorexia nervosa, irritable bowel syndrome (IBS), obsessive–compulsive disorder (OCD), Parkinson's disease). Query 1: AREA[InterventionSearch](psilocybin OR psychedelic OR LSD OR GM2505 OR GH001 OR BPL-003 OR RE02 OR BMND08 OR IBX-210 OR ELE-101 OR TRYP-880 OR PEX010 OR SYNP-101 OR MSP-1014 OR APEX-90 OR APEX-52 OR MM120 OR MB22001 OR CYB0004 OR RE104) NOT AREA[InterventionSearch](cannabis OR dronabinol OR cannabinoid OR MDMA OR ketamine) AND AREA[OverallStatus](RECRUITING) AND AREA[Phase](PHASE3 OR PHASE2). Query 2: AREA[InterventionSearch](MDMA OR methylone OR methylenedioxymethamphetamine OR midomafetamine) NOT AREA[InterventionSearch](cannabis OR dronabinol OR cannabinoid) AND AREA[OverallStatus](RECRUITING) AND AREA[Phase](PHASE3 OR PHASE2).

sex (male/female)? Can plasticity signatures be used for patient stratification or prognostication? Do transformational or mystical experiences in the absence of psychedelic drugs produce similar neurotrophic effects to psychedelics? Can psychedelics stimulate the neurotrophic pathway if the subjective experience is prevented?

Behavioral change and the plasticity window

Increased psychedelic-induced plasticity may manifest as increased behavioral or psychological flexibility. In rodent models, psychedelics seem to increase behavioral flexibility across multiple paradigms. Ketamine and psychedelics can accelerate fear extinction (the unlearning of negative associations that are no longer valid) in learning paradigms considered to be relevant to post-traumatic stress disorder[94]. However, the timing of behavioral interventions relative to psychedelic exposure may be critical. For instance, psilocybin enhanced fear extinction only when extinction training occurred within a specific post-drug window. Similarly, psychedelics reduced heroin self-administration in rats when reexposure to drug-associated cues was timed to coincide with heightened plasticity[78]. And particular social learning paradigms have found that psychedelics open a window of increased social reward learning, but the duration of that window varies between drugs[95]. Even in the relatively well-differentiated visual system, ketamine and psychedelics may transiently restore plasticity and modulate visual adaptation[96–98].

These findings support the idea that psychedelics open a transient window during which experience and environment can direct lasting behavioral change. This window has parallels to critical periods and sensitive periods in neurodevelopment[99,100]. The possibility of applying these observations to the clinical use of psychedelics is enticing[99,101].

Therapy in the plasticity window

There is growing interest in the idea that psychedelics may open a window of psychological plasticity that can be leveraged therapeutically[99]. This is conceptualized as a period of time when individuals have a greater ability to restructure maladaptive patterns of thought and behavior associated with psychiatric illness. During this time, clinicians may help patients to integrate insights from their psychedelic experience or adopt behavioral changes[99,102–106]. Individuals do show increases in openness—a personality trait linked to flexible information processing—following psychedelic therapy[46,107].

This imparts particular importance to therapeutic 'integration' sessions after dosing. Indeed, subjective measures of psychological insight and meaning-making processes during the integration period predict lasting benefit[108]. Effective integration may allow patients to consolidate new emotional and behavioral responses while the brain is still in a plastic state.

Experts in the field have widely differing perspectives on the type and amount of psychological support that is optimal[109]. Ongoing clinical work is exploring how different contextual manipulations and permutations of existing therapeutic approaches might reduce the likelihood of a 'bad trip' and optimize therapeutic outcomes. Most clinical trials published thus far have manipulated the drug treatment condition while holding the non-drug aspects of the treatment model constant. A recent 2 × 2 factorial study of ketamine therapy for relapse prevention in alcohol use disorder found that both drug and psychotherapy had independent and additive benefits on outcome[110]. An older study with LSD found conflicting results[111]. However, the more directive psychotherapies used in that study (for example, hypnodelic therapy) are no longer considered effective. Further data are needed to establish an empirical basis by which to estimate the benefit of various forms of therapy and support.

Clinical development

Overall, the clinical development of psychedelics has moved from small proof-of-concept studies toward large-scale phase 3 trials across multiple indications, reviewed in the sections below. The strongest

evidence currently supports the use of classic psychedelics—delivered with psychological support—for mood disorders, with promising but less mature evidence for anxiety, addiction, pain-related disorders and other neuropsychiatric conditions. As the field progresses, challenges related to trial design, stigma, policy, cost, scalability, integration and regulatory pathways must be systematically addressed to ensure that psychedelic therapies can be safely and effectively translated into mainstream clinical practice.

Mood and anxiety disorders

Clinical research with psychedelics has made the greatest progress in relation to depression, particularly major depressive disorder and treatment-resistant depression[112]. Psilocybin has demonstrated rapid and sustained antidepressant and anxiolytic effects across several phase 2 randomized controlled trials[3,6,7] (with comparators being placebo, SSRI or lower doses of psilocybin), earning Breakthrough Therapy designation from the US Food and Drug Administration (FDA). Smaller trials also suggest antidepressant potential for ayahuasca[113] and LSD[114]. Several phase 2/3 studies from industry sponsors and academia are underway (Fig. 3) to confirm/extend these findings and support regulatory approval.

Emotional blunting and anhedonia may be an area of clinical opportunity. First-line treatments for depression (for example, SSRIs) often lift low mood at the cost of blunting emotions. In a head-to-head comparison ($N = 59$), psilocybin and the SSRI escitalopram showed comparable efficacy for depression, but psilocybin was superior in treating anhedonia and emotional deficits in individuals with depression[3,115]. While the SSRI blunted emotion, psilocybin enabled participants to experience and tolerate stronger emotions. This result was replicated in a larger trial ($N = 233$) of psilocybin in depression[6,116].

Studies of psychedelics for depression or existential distress in individuals with terminal illness[8,12] have also consistently shown secondary benefits for anxiety. Two recent placebo-controlled trials evaluating LSD specifically for anxiety disorder reported significant reductions in anxiety symptoms at 4 weeks[11] and 16 weeks[9]. Phase 3 trials of LSD for generalized anxiety disorder are underway (Fig. 3).

While the growing body of evidence supporting psychedelics for mood disorders is compelling, important clinical questions remain, relating to the durability of therapeutic effects, optimal re-dosing strategies, and the identification of biomarkers predictive of individual response. Ongoing phase 3 trials with continuation and re-dosing procedures (for example, NCT06308653, NCT05711940 and NCT05624268) will help answer some of these questions. Methodological issues, such as expectancy effects, unblinding and heterogeneity in the type and dose of psychotherapeutic support, complicate interpretation of the current literature (see 'Expectancy and unblinding').

Addiction

Studies into the potential of psychedelics to treat substance use disorders have been ongoing since the 1950s. Preclinical studies support possible anti-addictive properties of psychedelics, and clinical trials have suggested safety, feasibility and potential therapeutic benefit of several different psychedelics[117]. A meta-analysis of historic studies suggested efficacy of LSD-assisted therapy for alcoholism[118]. This finding was supported by a recent randomized controlled trial showing that two psilocybin sessions significantly reduced alcohol consumption in community-dwelling individuals with alcohol use disorder[14]. In contrast, a smaller study in recently detoxified individuals found no significant benefit of a single psilocybin dose on time to relapse[119]. These studies differed in several important respects, including sample size, participant population, primary outcomes and number of dosing sessions. Ongoing studies of these and other psychedelics and non-psychedelic analogs will help determine replicability and assess how these factors contribute to efficacy (Fig. 3).

Ongoing research is evaluating the effects of psychedelics to treat several other substance use disorders. Psilocybin has shown promising results in a pilot study for smoking cessation[120], prompting an ongoing multisite trial (NCT05452772). Trials are also ongoing to evaluate the effects of psilocybin in cocaine (NCT06102434) and methamphetamine (NCT04982796) addiction.

Beyond the classic psychedelics, ibogaine—a complex, polypharmacological compound with psychedelic-like psychoactive properties (Table 1)—has been of particular interest for the treatment of opioid addiction[121] (and traumatic brain injury[122]). However, it has not yet been widely studied in formal trials due to long duration of acute psychoactive effects (lasting 12–24 h) and safety concerns, as Na^+ channel inhibition produces arrhythmia risk[123,124] and the metabolite nor-ibogaine prolongs QTc interval for days[125].

Overall, questions remain about how best to apply psychedelics in the treatment of addiction. Beyond identifying which psychedelics are effective for which substance use disorders, there is a wide range of variables that still need to be explored. Key unknowns include the optimal timing of psychedelic therapy in the context of broader addiction treatment and the ideal dosing strategy (for example, whether a single session is sufficient or if multiple sessions may help sustain benefits).

Pain and opioid use

Chronic pain remains the leading cause of disability in the United States. Preclinical findings on psychedelics' analgesic properties have been mixed: some studies report no direct analgesic effects of psilocybin in rodent models[126], others have observed reductions in pain-related behaviors and hypersensitivity[127,128]. Psychedelics may improve quality of life, functionality and distress in people with chronic pain[129]. It should be noted that while recent progress has been made in understanding and treating nociceptive pain, chronic pain disorders have largely remained intractable. For this reason, multiple National Institutes of Health institutions have joined together (https://grants.nih.gov/grants/guide/rfa-files/RFA-AG-25-004.html) to fund a wider exploration of the translational and clinical potential of psychedelics for chronic pain.

PTSD

In the treatment of PTSD, the primary focus has been on 3,4-methylenedioxy-methamphetamine (MDMA)-assisted therapy—although trials of psilocybin and other psychedelics are also underway. Phase 3 trials conducted by the Multidisciplinary Association for Psychedelic Studies reported that MDMA, combined with extensive psychotherapy, produced large and durable reductions in PTSD symptoms[130,131]. However, the FDA declined to approve MDMA for PTSD. Despite large effect sizes, an advisory committee meeting raised several concerns regarding trial design, integrity and bias. At present, the path forward for MDMA for PTSD in the United States remains unclear. However, preliminary phase 2 trial results suggest that the empathogen 3,4-methylenedioxy-*N*-methylcathinone (methylone) demonstrated an efficacy signal comparable to that of MDMA[132].

By contrast, the use of classic psychedelics in PTSD has not received as much focus. Translational neuroscience studies indicate that psilocybin can produce shifts in affective processing (including decreases in amygdala responses to negative emotional stimuli)[133–135], which suggests potential in the treatment of PTSD. These findings warrant clinical assessment, and phase 2 testing is underway (NCT06888128). Establishing specialized preparation and integration frameworks tailored for trauma-focused therapy may be necessary to ensure efficacy and safety.

Other psychiatric and neuropsychiatric conditions

Beyond the above indications, psychedelics have been or are currently being investigated for obsessive–compulsive disorder[136], anorexia nervosa[137], body dysmorphic disorder[138], chronic pain disorders[129,139]

and existential distress in individuals with terminal illnesses[8,12]. Data from observational studies and case series suggest psilocybin and LSD may alleviate cluster headaches[140] and migraine[141], and ibogaine may benefit veterans with traumatic brain injury[122]. Several small clinical trials are underway to evaluate these effects more rigorously (for example, NCT06399263, NCT06518720, NCT06731335, NCT06001749, NCT05351541, NCT03781128 and NCT05477459). A challenge in these emerging indications is the difficulty in distinguishing psychological benefits (such as increased acceptance or reduced catastrophizing) from direct neurophysiological effects (for example, pain processing). Clearer mechanistic studies and careful trial designs will be critical for expanding the therapeutic footprint of psychedelics into these areas.

Challenges and opportunities for progress

Despite rapid scientific progress, psychedelic treatments pose serious practical, methodological and regulatory challenges. Chief among these challenges are the unmistakable consciousness-altering effects of psychedelics, the possible role of these effects as mediators of therapeutic benefit, the likelihood of clinically significant interactions between the direct effects of the drug and the therapeutic setting, and a cultural environment that is characterized by exaggerated expectations, both positive and negative, regarding the benefits and harms of psychedelics. While none of these individual issues are unique to psychedelics, in combination they have forced clinical scientists, drug developers and regulatory bodies to develop new approaches for the study of psychedelic treatments.

Expectancy and unblinding

Multiple factors contribute to the expectation of study participants (and study clinicians) that psychedelic treatment will benefit them more than the alternative[142]. When combined with the intense and unmistakable nature of the psychedelic experience, this can undermine the rigor of standard double-blind trial designs[143]. Often, greater than 90% of participants correctly identify whether they received an active psychedelic (functional unblinding)[14]. Because of this, some have questioned the interpretability of trial results[144]. A recent meta-analysis comparing trials of psychedelics to trials of typical antidepressants supports the legitimacy of these concerns[145]. While it is true that psychedelics produce larger drug–placebo separation than typical antidepressants, much of the difference comes from a smaller placebo response in psychedelic trials than typical antidepressant trials, rather than a larger response to active treatment.

The FDA has advised both academic and industry sponsors to more rigorously assess blinding integrity and manage expectancy bias in psychedelic trials[146]. One proposed approach is to compare psychedelics against other psychoactive agents, such as cannabis, to improve active blinding[147]. Dose–response designs and open-label comparative efficacy studies may provide important evidence along with gold-standard blinded randomized controlled trials. Ultimately, a plurality of (imperfect) solutions may be needed to tackle this complex challenge of attributing the cause of therapeutic outcomes in trials of psychedelics[148].

The role of set, setting and psychotherapy

The role of context—or 'set and setting'—in shaping psychedelic experiences is well recognized[149], although not unique to psychedelics. Recent studies have shown that environmental and task-related factors can strongly modulate brain activity, therapeutic effects and safety. For example, two mechanistic clinical studies using magnetoencephalography and fMRI found that greater sensory stimulation and cognitive demands dampened psychedelic-induced brain changes[51,150]. Remarkably, interrupting the internal psychedelic experience by asking participants to rate drug intensity every minute reduced both the subjective and antidepressant effects of *N*,*N*-dimethyltryptamine (DMT)[151]. In line with this, individuals under the influence of a psychedelic during a mass trauma event reported that psychedelic effects abruptly subsided, and then returned once safety was reached[152].

Relatedly, the type of psychotherapy that should be paired with psychedelic treatment remains contested. Many consider psychotherapy a critical or even primary ingredient—as in 'psychedelic-assisted therapy'—while others view the psychedelic itself as the treatment[153]. Like other psychotherapeutic interventions, the alliance and rapport between therapist and study participant seems to moderate responses to psilocybin for depression[108,154,155]. Certain aspects—such as the three-stage model of preparation (before dosing), support and integration (after dosing), and an inward-directed approach—have been consistent across recent psychedelic clinical trials. But the frequency and dose of psychotherapy or psychological support can vary across psychedelic trial facilitators and studies[109]. Ongoing phase 3 trials of psychedelics differ considerably in the amount and type of psychotherapy (NCT06308653, NCT05711940 and NCT05624268). Results of these studies will provide some clues as to the primacy of psychotherapy both to enhance efficacy and mitigate risk[156]. It is possible that the role of psychotherapy and/or clinical support may vary depending on the clinical indication, participant characteristics and treatment setting.

An expanding drug class

The majority of psychedelic clinical trials in recent years have used psilocybin. But a closer look reveals psychedelics to be a rich and growing class of compounds. There are at least six naturally occurring serotonergic psychedelic compounds, more than 100 psychedelic compounds currently under development by pharmaceutical companies[157] and many additional synthetic compounds with documented psychedelic or hallucinogenic effects in humans—many first described and systematically cataloged in the PiHKAL and TiHKAL volumes[158,159]. AI-based tools for drug discovery (that is, AlphaFold2) suggest that the number of unique molecules with psychedelic 5-HT_{2A} agonism could be over 100,000 (refs. 160,161). These compounds may differ not only in the profile of receptors that are co-activated, but also in the ways in which the 5-HT_{2A} itself is engaged and activated[162]. The number of possible combinations between drug, dosing protocol and medical indication are limitless. Thus far, attempts to match specific psychedelics to specific indications (for example, ibogaine for opioid use disorder) are largely speculative.

Risks associated with psychedelics

According to a recent analysis, across >2,000 modern participants exposed to psilocybin, LSD, DMT or MDMA (all under clinical supervision), there were no deaths or psychosis-related hospitalizations attributable to the study drug[163]. Psilocybin and LSD have a low risk of addiction, physiological toxicity or overdose[163–165]. However, concerns remain regarding impaired perception and judgement, adverse mental health outcomes, such as anxiety and derealization[166,167], and long-lasting perceptual and cognitive disruptions after use. Notable adverse reactions in recent large trials of psilocybin have included psychological discomfort (~30%), physical discomfort and anxiety during the session[6,7,11]. Although rare, increased suicidal ideation or behavior has been observed following high-dose psilocybin in several clinical trials[6,168,169] and case reports[170,171], contrasting with evidence of anti-suicidal effects of psilocybin-assisted psychotherapy[172].

Of note, some psychedelics are far more toxic than psilocybin or LSD (for example, Bromo-DragonFLY, NBOMe series compounds, and others) due to α-adrenergic agonism or hERG channel blockade that triggers vasoconstriction, hyperthermia or arrhythmia—often compounded by slowed monoamine-oxidase or CYP metabolism, increasing the risk for seizures or serotonin syndrome.

The resurgence of interest in psychedelics has been paralleled by rising recreational use[173], presenting a set of risks and adverse events that differ somewhat from those seen in clinical settings[174,175].

Therapeutic use of psychedelics in clinical trials generally relies on one or two macrodoses—doses sufficient to generate the full psychedelic experience—administered weeks apart. In contrast, microdosing—chronic, repeated exposure to low doses—has gained popularity outside the medical establishment. This pattern of use may engage different pharmacological pathways and carries theoretical cardiovascular risks, particularly due to the potential cumulative effects of 5-HT_{2B} receptor activation[176,177]. Chronic stimulation of 5-HT_{2B} receptors causes pathological proliferation of cardiac valve fibroblasts, leading to valvulopathy. This mechanism is well established from drugs such as fenfluramine (Fen-Phen), pergolide and methysergide, which were withdrawn from the market after approximately 25% of individuals taking these medications daily developed valvular heart disease[178–180]. A recent preclinical study in mice found no cardiovascular pathology after chronic microdosing with LSD, attributing this to lower levels of 5-HT_{2B}-receptor occupancy with LSD than known cardiotoxins such as fenfluramine[181]. However, research on cardiovascular risks of microdosing in humans remains limited[182].

Hallucinogen persisting perception disorder is defined in the Diagnostic and Statistical Manual of Mental Disorders, 5th Edition, as persistent visual distortions following psychedelic use. While the mechanism is unclear, theories speculate that lasting disruption to visual processing may be mediated by GABAergic disinhibition or thalamocortical dysregulation[183]. But it appears that there is a broader range of potential perceptual and cognitive disturbances that can last for weeks or months, including dissociation, depersonalization and a troubling sense of psychological disconnection[184]. Risk factors for persistent adverse effects include young age[185], a history of psychiatric illness, unsafe or unsupported use, lack of therapeutic integration and possible genetic vulnerability[186].

A related concern is whether psychedelics can precipitate schizophrenia spectrum disorders in vulnerable individuals[187–189]. Population studies have found mixed evidence of elevated risk[190–192]. A correlational study based on a representative national (US) survey found lower rates of psychiatric illness in those who had used psychedelics versus non-psychedelic users, after controlling for other substances and covariates[193].

These adverse psychological or psychiatric effects may be consistent with the plasticity-inducing effects of psychedelics. A sudden increase in psychological and behavioral malleability likely poses inevitable risks, particularly in the absence of a safe and supportive environment[194]. The same action that can promote transformational change (namely, enhanced plasticity) may underlie the greatest risks of psychedelics. This theory has important implications regarding the ability to separate therapeutic benefit from inherent risks. It seems to clash with recent evidence (discussed in the following section) that it may be possible to decouple therapeutic mechanisms from intense alterations in consciousness.

NHPAs

The development of NHPAs stems largely from preclinical work suggesting that psychedelics' neuroplastic effects can be separated from their hallucinogenic effects[78,195,196]. Compounds such as 2-Br-LSD, AAZ-A-154, ITI-1549 and tabernanthalog appear to engage the 5-HT_{2A} receptor in a biased or functionally selective manner, preserving downstream signaling linked to plasticity (for example, BDNF/TrkB activation, dendritic growth) while avoiding behavioral correlates of hallucinations, such as the head-twitch response in rodents[197,198]. Still, the relevance of these findings to humans remains uncertain.

Participants in clinical trials often describe their experiences as emotionally intense and personally meaningful—and many attribute subsequent improvements in mood, behavior or worldview to these moments of insight or catharsis[199]. In trials of classic psychedelics, the quality and intensity of the subjective experience—particularly mystical-type effects, and emotional breakthroughs—consistently correlate with positive clinical outcomes[200]. Yet recent discoveries have challenged the assertion that perceptual and experiential effects of psychedelics are necessary for their therapeutic benefits.

Parallels are sometimes drawn to ketamine (Table 1), an *N*-methyl-D-aspartate receptor antagonist that also produces psychoactive effects and induces rapid neuroplastic and antidepressant effects. Despite decades of study, the relationship between subjective effects and antidepressant response remains disputed. Some studies have found an association between the two[201], others have reported that ketamine can remain effective at a dose with minimal dissociative effects[202–204] or with 'safeners' that reduce these effects[205,206]. A recent attempt tested this relationship by administering ketamine during general anesthesia[207]. Treatment guesses revealed that blinding was successful, yet robust antidepressant responses were observed across all treatment groups. The authors interpret the overall pattern as consistent with the idea that expectancy, salient experience (that is, surgery) and other nonspecific perioperative factors can drive depression symptom change even when blinding is maintained.

At this stage, the importance of the subjective experience and the clinical potential of NHPAs remain unresolved questions. It is possible that the experiential and plasticity mechanisms are partially dissociable, with different compounds offering different therapeutic advantages. It may also be that more elaborate or emotionally salient experiences—whether psychedelic or not—boost psychological change in the short term, even if plasticity is the primary driver of lasting improvement. Either way, NHPAs offer a powerful scientific tool for probing the contributions of receptor-level pharmacology, neural plasticity and conscious experience to psychiatric treatment. Whether they evolve into therapeutics themselves—or primarily clarify the mechanisms of psychedelic action—remains to be seen.

Scalability and infrastructure

Psychedelic therapy is a radical departure from the conventional pharmacotherapy paradigm in psychiatry[1]. A further obstacle to widespread implementation of psychedelic therapy is the resource-intensive nature of the intervention. Most protocols require several hours of preparatory and integration sessions in addition to the dosing experience itself, often involving two trained facilitators per participant. This raises questions about how to scale psychedelic medicine while maintaining therapeutic quality and safety. Group-based models[208,209], digital integration support and hybrid telehealth approaches may help address these constraints. The ongoing move toward interventional psychiatry in the ambulatory setting will help make these hurdles more surmountable. Clear regulatory guidance and reimbursement policies will be essential to ensure equitable access while maintaining high standards of safety and efficacy.

Global regulatory landscape

Globally, the regulation of psychedelics varies considerably[210]. For example, Canada permits physician-supervised psychedelic use under the Special Access Program[211]; Australia recently reclassified psilocybin and MDMA to allow prescribing for treatment-resistant depression and PTSD under strict controls[212]; and countries such as the Netherlands, Switzerland and Israel maintain more permissive pathways for clinical research or limited therapeutic use. Other countries have taken distinctive approaches: Portugal has decriminalized personal possession of all drugs, including psychedelics, shifting toward harm reduction frameworks rather than criminal penalties; Brazil permits certain indigenous and religious uses of ayahuasca[213].

Under US federal law, psychedelics remain schedule I prohibited substances. This has not stopped several states and municipalities from enacting laws to decriminalize or legalize certain psychedelics for medical or religious use[214,215]. These initiatives have complicated the development of psychedelic treatments by blurring the lines between medical use of psychedelics within the healthcare system and other

categories of use. If psychedelic therapies are approved by the FDA, issues related to drug rescheduling, medical/therapeutic oversight, cost, public health messaging and societal impacts will require careful consideration. By policy, the FDA does not evaluate the efficacy of psychotherapy. So, it remains to be seen what requirements and restrictions may be placed on psychedelic-assisted psychotherapy, should psychedelics receive regulatory approval.

Conclusion

Psychedelic science has made remarkable strides, developing from fringe curiosity to a frontier of clinical neuroscience. The growing body of evidence supporting the therapeutic potential of psychedelic therapy to improve outcomes in mood, trauma-related disorders and substance use disorders is both compelling and complex. It raises fundamental questions about mechanisms, treatment paradigms, safety, regulation and equitable access. Psychedelics represent a paradigm shift in how we conceptualize and intervene in mental illness in Western medicine. The psychedelic paradigm places experience, neurobiology and environment in dynamic interplay, more in keeping with a biopsychosocial framework than a purely biological one. Whether through psychological insight, targeted molecular design or both, the future of psychedelic psychiatry will depend on how we balance innovation with therapeutic responsibility. Deepening our understanding of the neurobiological mechanisms underlying these effects will be essential not only to refine treatments, but also to ground this rapidly evolving field in rigorous, reproducible science.

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Author contributions

All authors contributed to outlining the ideas and to writing and revising the manuscript. J.S.S. conducted analyses of clinical trials, reviewed drug classes and assembled the figures and tables.

Competing interests

J.S.S. is a consultant for Otsuka and Janssen and holds a provisional patent for the use of precision functional mapping for measuring target engagement by experimental therapeutics. C.L. has served as a scientific advisor to Delix Therapeutics and Brainify. G.E.N. has served as principal or co-investigator on studies funded by Alkermes, LB Pharmaceuticals, COMPASS Pathways, Usona Institute and Reunion Neuroscience. G.E.N. has served as a paid consultant for Carelon and Alkermes. R.L.C.-H. is a scientific advisor to TRYP therapeutics, MindSpace and Otsuka. M.P.B. has received research funding from Tilray Canada, Lykos Therapeutics, B.More, Ceruvia Life Sciences and Mind Medicine, serves on the Advisory Boards of Ajna Labs, Journey Colab and Bright Minds Biosciences, and provides consultation to Lilly USA and Otsuka Pharmaceutical Development & Commercialization.

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